

MSc Project 1: Time-Series-Enhanced Predictive Process Monitoring Phase Review 4

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Abstract. This document reviews the Testing and Assessment phase of the project. Building on the infrastructure completed in Sprint 3, this phase ran the full experimental comparison between log-only and time-series-enhanced models for both classification and regression tasks on the complete BPI Challenge 2017 dataset, implemented explainability and diagnostic tooling, and produced the empirical results that answer the project's central research question. The phase concludes with all experimental findings documented and the final report as the remaining deliverable.

Keywords: Predictive Process Monitoring · Model Evaluation · Feature Importance · Time-Series Features · Overfitting Analysis

1 Introduction & Objectives

1.1 Phase Goals

The Testing and Assessment phase aimed to run the full experimental suite on the complete BPI Challenge 2017 dataset [1], moving beyond the development subset used during earlier sprints. We compared log-only and time-series-enhanced models for both outcome classification and remaining-time regression under identical conditions, producing the head-to-head comparison deferred from Sprint 3. We also built diagnostic and explainability tooling: permutation feature importance, SHAP summary plots, overfitting analysis, confusion matrices, and predicted-versus-actual scatter plots to interpret the models and assess their reliability.

1.2 Team Organization

Work continued to be tracked through GitHub Issues. Michał drove most of the experimental pipeline: model comparisons for both classification and regression, confusion matrices, overfitting metrics, and feature importance data. Konstantinos built the SHAP explainability module, added binary outcome classification, refactored the evaluation metrics for the final report, and wrote the

architecture documentation. Roman handled class-balanced training with cross-validation search, implemented the baseline models, and fixed the prefix generation logic for the classification task. All team members contributed to refining the empirical findings.

1.3 Role Allocation

Role assignments from Sprint 3 carried forward into this phase, with all team members contributing across tracks as needed.

Table 1. Role assignments for the Testing and Assessment phase.

Role	Assigned To
Project Coordinator / Scrum Master	Roman
Technical Lead / Architecture	Konstantinos
Data & Preprocessing Lead	Roman
Modelling & Experimentation Lead	Michal
Evaluation & Visualization Lead	Konstantinos
Quality & Tooling (Tests, CI)	Roman
Documentation & Reporting	Michal, Roman, Konstantinos (shared)

2 Technical Implementation

This phase did not introduce new architecture. We ran the existing infrastructure from Sprints 1–3 at full scale on the complete BPI Challenge 2017 dataset. Table 2 maps each work item to its issues and pull requests.

2.1 Full-Scale Experimental Runs

We ran all experiments through the parametric CLI entry point from Sprint 2. For classification we compared four models: Dummy, Logistic Regression, Random Forest, and HistGradientBoosting using the full 58-feature log+time-series set. For regression Ridge replaced Logistic Regression in the same four-model design. All configurations shared the 0.8-quantile temporal train/test split by case start time, prefix lengths from 3 to 100, and identical random seeds. Hyperparameter search used `RandomizedSearchCV` with 3-fold cross-validation and 10 iterations. HGB came out best for classification; Random Forest for regression.

2.2 Time-Series Impact Analysis

For the central comparison we ran the best model from each task under both configurations with all other settings held constant. Hourly-resampled time-series features with 8-hour rolling windows did not improve classification and degraded regression performance compared to the log-only baseline. Unbounded prefixes and capped prefixes produced nearly identical results.

2.3 Diagnostic and Explainability Tooling

We ran several diagnostic analyses to interpret the models. Permutation feature importance ranked activity-count features at the top for both tasks. SHAP summary plots visualised how individual feature values push predictions toward specific outcomes. Overfitting looked moderate: train-test gaps stayed within expected bounds for tree-based ensembles. We also produced confusion matrices and predicted-versus-actual scatter plots for every model-task combination.

Table 2. Testing and Assessment implementation steps.

Step	Description	Issues / PRs
1	Run experiments for both tasks	#120
2	Regression model comparison	#119
3	Classification model comparison with HGB	(#112-#118)
4	Confusion matrices	—
5	Overfitting metrics	—
6	Feature importance to CSV	—
7	Predicted-vs-actual scatter plots	—
8	HGB as best classification model	—
9	Full-prefix-length configuration	—
10	Test suite maintenance and fixes	—

3 Review & Retrospective

3.1 Successes

All planned phase objectives were completed. The full experimental suite ran end-to-end on the complete BPI Challenge 2017 dataset, producing the head-to-head comparison deferred from Sprint 3. The diagnostic tooling ran without issues and its results fed into the interpretation. Table 3 lists the key issues and pull requests closed during the phase.

Table 3. Testing and Assessment issues and pull requests.

Issue / PR Description	
#120	Run experiments for both tasks
#119	Compare regression models
#112-#118	Classification experiments with HGB
–	Comput confusion matrices
–	Evaluate overfitting metrics
–	Save feature importance to CSV
–	Predicted vs actual scatter plots
–	HGB as best model for classification
–	Add config for full prefix len
–	Fix tests (remove integration tests)

3.2 Difficulties

The main challenge was the negative result itself: the time-series features added no value for classification and degraded regression. Explaining this outcome required deeper diagnostic analysis than initially scoped. The key insight was concept drift: the time-series features are global process-state signals whose distributions shift between train and test on the chronological split, so they fail to generalise. We also investigated proxy effects, extraction-template limitations, and the curse of dimensionality. The discovery of the `WaitingStateFeatures` activity-matching bug, where matching on `0_Sent` silently never fired because the real event names include parenthetical suffixes, occurred during this phase and needed to be documented as a limitation. Right-censoring at the end of the log surfaced only during long-prefix analysis. All findings went into the evaluation discussion.

3.3 Task Tracking

We managed all work through GitHub Issues with linked pull requests. The requirements tracking system kept detailed sub-requirements as GitHub issues linked to parent requirements instead of editing the requirements document, which worked well for scope changes like adding HGB as a model family.

4 Moving Forward

4.1 Future Planning

With the Testing and Assessment phase complete, the remaining deliverable is the final report. We will produce the document covering data preprocessing, feature engineering, model specification, experimental protocol, and results. It will include a critical evaluation of the time-series features, limitations, and a project retrospective. All remaining open issues will be closed and the repository documentation finalised alongside.

4.2 Lessons Learned

The main lesson from this phase is that building extraction machinery is not the same as validating that the features carry useful signal. We put more effort into constructing the pipeline than into checking whether the extracted signals tracked real workload variation, and the null result reflects that imbalance. Claims about time-series features also need to be scoped to the specific extraction strategy rather than generalised. These are observations we carry forward into the final report.

References

1. van Dongen, B.F.: BPI Challenge 2017. Eindhoven University of Technology (2017). <https://research.tue.nl/en/datasets/bpi-challenge-2017>